MATHS BRIDGE
ACADEMY
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SECTION A — 1 MARK EACH ($9 \times 1 = 9$)

Q1. HCF and LCM [1 mark]

HINTS / SOLUTION	MARKS
HCF $\times$ LCM = product of numbers. $\therefore$ LCM $= (306 \times 657) \div 9 = 22{,}338.$	1 (no part marks)

Q2. Number of zeroes from graphs [1 mark]

HINTS / SOLUTION	MARKS
Number of intersections with x − axis: (i)0; (ii)1; (iii)3; (iv)1; (v)4; (vi)5.	1 (no part marks)

Q3. Arithmetic progression [1 mark]

HINTS / SOLUTION	MARKS
$78 = 3 + (n - 1)5 \Rightarrow 75 = 5(n - 1) \Rightarrow n = 16.$ $\therefore$ 16th term.	1 (no part marks)

Q4. Distance formula [1 mark]

HINTS / SOLUTION	MARKS
$\therefore d = \sqrt{[(4 - 2)^2 + (1 - 3)^2]} = \sqrt{8} = 2\sqrt{2}$units.	1 (no part marks)

Q5. Similarity of shapes [1 mark]

HINTS / SOLUTION	MARKS
(i)similar; (ii)similar. Both blanks must be correct for the 1 mark (no part marks).	1 (no part marks)

Q6. Length of tangent [1 mark]

HINTS/SOLUTION	MARKS
OP $\perp$ PQ. PQ $= \sqrt{OQ^2 - OP^2} = \sqrt{12^2 - 5^2} = \sqrt{119}$cm. (D).	1 (no part marks)

Q7. Trigonometric expression [1 mark]

HINTS/SOLUTION	MARKS

$\frac{2\tan 30°}{1+\tan^2 30°} = \frac{2\left(\frac{1}{\sqrt{3}}\right)}{1+\frac{1}{3}} = \frac{\sqrt{3}}{2} = \sin 60°$. (A).	1 (no part marks)

Q8. Area of sector [1 mark]

HINTS/SOLUTION	MARKS
Area $= \left(\frac{p}{720}\right) \times 2\pi R^2$. (B).	1 (no part marks)

Q9. Probability range [1 mark]

HINTS/SOLUTION	MARKS
Probability must lie between 0 and 1 inclusive. -1.5 is impossible. (B).	1 (no part marks)

SECTION B — 2 MARKS EACH ($15 \times 2 = 30$)

Q10. Composite numbers [2 marks]

HINTS/SOLUTION	MARKS
$7 \times 11 \times 13 + 13 = 13(7 \times 11 + 1)$ $= 13 \times 78$; both factors exceed 1, so composite.	1
$7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ $= 5(7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1)$ $= 5 \times 1009$; composite.	1

Q11. Zeroes and coefficient relations [2 marks]

HINTS/SOLUTION	MARKS
$x^2 - 2x - 8 = (x-4)(x+2)$; zeroes $\alpha = 4, \beta = -2$.	1
$\alpha + \beta = 4 - 2 = 2 = -\frac{b}{a}$; $\alpha\beta = 4(-2) = -8 = \frac{c}{a}$. Both verified.	1

Q12. Quadratic polynomial from sum and product [2 marks]

HINTS/SOLUTION	MARKS
Required polynomial: $x^2 -$ (sum of zeroes)$x +$ product of zeroes.	1
$= x^2 - \frac{x}{4} - 1$; multiplying by 4 gives $4x^2 - x - 4$.	1

Q13. Consistency of linear equations [2 marks]

HINTS/SOLUTION	MARKS
(i) $\frac{a^1}{a^2} = \frac{3}{2}$ and $\frac{b^1}{b^2} = \frac{2}{-3} = -\frac{2}{3}$;	1

unequal $\Rightarrow$ unique solution, consistent.	
$(ii)\frac{a^1}{a^2}=\frac{2}{4}=\frac{1}{2},\frac{b^1}{b^2}=\frac{-3}{-6}=\frac{1}{2},\frac{c^1}{c^2}=\frac{-8}{-9}=\frac{8}{9};$ parallel distinct lines $\Rightarrow$ inconsistent.	1

Q14. Nature of quadratic roots [2 marks]

HINTS/SOLUTION	MARKS
For $2x^2-3x+5=0: a=2, b=-3, c=5;\ D=b^2-4ac.$	1
$D=9-40=-31<0\ \therefore$ no real roots.	1

Q15. AP term equal to zero [2 marks]

HINTS/SOLUTION	MARKS
$a+2d=4;\ a+8d=-8$ $\Rightarrow 6d=-12$ $\Rightarrow d=-2;\ a=8.$	1
$a_n=8+(n-1)(-2)=0$ $\Rightarrow n-1=4$ $\therefore n=5.$	1

Q16. Common difference [2 marks]

HINTS/SOLUTION	MARKS
$a_{17}=a+16d$ and $a_{10}=a+9d.$	1
$a_{17}-a_{10}=7$ $\Rightarrow 7d=7$ $\therefore d=1.$	1

Q17. Section formula [2 marks]

$\frac{\text{HINTS}}{\text{SOLUTION}}$	MARKS
$(AP:PB=2:3), P=\left(\frac{2x_2+3x_1}{2+3},\frac{2y_2+3y_1}{2+3}\right).$	1
Substituting$(A(-1,7)), (B(4,-3),$ $\therefore P=\left(\frac{2\times4+3\times(-1)}{5},\frac{2\times(-3)+3\times7}{5}\right)=(1,3).$	1

Q18. Basic proportionality theorem [2 marks]

$\frac{\text{HINTS}}{\text{SOLUTION}}$	MARKS
$(i)DE \parallel BC \Rightarrow \frac{AD}{DB}=\frac{AE}{EC}\Rightarrow\frac{1.5}{3}=\frac{1}{EC}\therefore EC=2$ cm.	1
$\frac{(ii)AD}{DB}=\frac{AE}{EC}\Rightarrow\frac{AD}{7.2}=\frac{1.8}{5.4}\therefore AD=2.4$ cm.	1

Q19. Radius using tangent [2 marks]

HINTS / SOLUTION	MARKS
Radius OT is perpendicular to tangent AT. $OA^2 = OT^2 + AT^2$.	1
$5^2 = r^2 + 4^2 \Rightarrow r^2 = 9 \therefore r = 3$ cm.	1

Q20. Cosine and tangent [2 marks]

HINTS / SOLUTION	MARKS
$\sin A = \frac{3}{4}$; $\cos^2 A = 1 - \sin^2 A$ $= 1 - \frac{9}{16} = \frac{7}{16}$ $\Rightarrow \cos A = \frac{\sqrt{7}}{4}$ (for acute A).	1
$\therefore \tan A = \frac{\sin A}{\cos A} = \frac{\frac{3}{4}}{\frac{\sqrt{7}}{4}} = \frac{3}{\sqrt{7}} = \frac{3\sqrt{7}}{7}$.	1

Q21. Height of tower [2 marks]

HINTS / SOLUTION	MARKS
$\tan 30° = \frac{\text{height}}{30} = \frac{1}{\sqrt{3}}$.	1
$\therefore$ Height $= \frac{30}{\sqrt{3}} = 10\sqrt{3}$m	1

Q22. Cone on hemisphere [2 marks]

HINTS / SOLUTION	MARKS
$V = \left(\frac{1}{3}\right)\pi r^2 h + \left(\frac{2}{3}\right)\pi r^3$; $r = h = 1$ cm.	1
$\therefore V = \frac{\pi}{3} + \frac{2\pi}{3} = \pi$ cm^3.	1

Q23. Probabilities on a die [2 marks]

HINTS / SOLUTION	MARKS
Sample space {1,2,3,4,5,6}; total outcomes = 6.	½
(i)prime {2,3,5} P(prime) $\Rightarrow \frac{3}{6} = \frac{1}{2}$.	½
(ii)strictly between 2 and 6 {3,4,5}	½

$\text{P(E)} \Rightarrow \frac{3}{6} = \frac{1}{2}.$	
(iii)odd {1,3,5} $\text{P(odd)} \Rightarrow \frac{3}{6} = \frac{1}{2}.$	½

Q24. Balls drawn from bag [2 marks]

HINTS / SOLUTION	MARKS
$\text{Total balls} = 3 + 5 = 8; \therefore \text{P(red)} = \frac{3}{8}.$	1
$\therefore \text{P(not red)} = \frac{5}{8}\left(\text{or } 1 - \frac{3}{8}\right).$	1

SECTION C — 3 MARKS EACH ($11 \times 3 = 33$)

Q25. Proof that $\sqrt{5}$ is irrational [3 marks]

HINTS / SOLUTION	MARKS
Assume $\sqrt{5} = \frac{a}{b}$, where a, b are co − prime positive integers and b $\neq 0$.	1
Squaring: $5 = \frac{a^2}{b^2} \Rightarrow a^2 = 5b^2$. Thus 5 divides a^2, hence 5 divides a. | Write $a = 5k$.	1
Then $25k^2 = 5b^2 \Rightarrow b^2 = 5k^2$, so 5 divides b too. Contradiction: a and b were co − prime. Hence $\sqrt{5}$ is irrational.	1

Q26. Cricket bats and balls (substitution) [3 marks]

HINTS / SOLUTION	MARKS
Let cost of a 1 bat = ₹x, cost of a 1 ball = ₹y. Then $7x + 6y = 3800$ … (1); $3x + 5y = 1750$ … (2).	1
From (2), $x = \frac{1750 - 5y}{3}$. Substitute in (1): $\frac{7(1750 - 5y)}{3} + 6y = 3800$ $\Rightarrow 12250 - 35y + 18y = 11400$ $\Rightarrow 17y = 850$ $\Rightarrow y = 50.$	1
$x = \frac{1750 - 5 \times 50}{3} = \frac{1500}{3} = 500.$ One bat costs ₹500 and one ball ₹50.	1

Q27. Right triangle sides [3 marks]

HINTS / SOLUTION	MARKS
Let base $= x$ cm; altitude $= (x - 7)$cm; hypotenuse $= 13$ cm.	½
By Pythagoras: $x^2 + (x - 7)^2 = 13^2 \Rightarrow 2x^2 - 14x - 120 = 0$ $\Rightarrow x^2 - 7x - 60 = 0.$	1
$(x - 12)(x + 5) = 0 \Rightarrow x = 12$ or -5; reject negative length.	1
$\therefore$ Base $= 12$ cm; altitude $= 12 - 7 = 5$ cm.	½

Q28. Classroom coordinate geometry [3 marks]

HINTS / SOLUTION	MARKS
From the graph: A(3,4), B(6,7), C(9,4), D(6,1)	1
AB $= \sqrt{[(6-3)^2 + (7-4)^2]} = \sqrt{18}$; BC $= \sqrt{[(9-6)^2 + (4-7)^2]} = \sqrt{18}$. Similarly CD $=$ DA $= \sqrt{18}$.	1
Diagonals AC $= 6$ and BD $= 6$. Four equal sides and equal diagonals establish a square. Champa is correct.	1

Q29. Trapezium diagonal ratio [3 marks]

HINTS / SOLUTION	MARKS
In ΔAOB and ΔCOD: $\angle AOB = \angle COD$ (vertically opposite). $\angle ABO = \angle CDO$ (alternate interior angles, AB ∥ CD).	1
$\therefore$ ΔAOB ~ ΔCOD (AA).	1
$\frac{AO}{CO} = \frac{BO}{DO}$ (C. s. s. t) $\Rightarrow \frac{AO}{BO} = \frac{CO}{DO}$, Hence proved.	1

Q30. Pole and tower shadows [3 marks]

HINTS / SOLUTION	MARKS
Since the sun's angle of elevation is the same, the two triangles are similar by the AA similarity criterion.	1
$\frac{\text{Height of pole}}{\text{Shadow of pole}} = \frac{\text{Height of tower}}{\text{Shadow of tower}}$,	1

$\frac{6}{4} = \frac{\text{Height of tower}}{28}$	
$\therefore$ Height of tower $= \frac{28 \times 6}{4} = 42$ m.	1

Q31. Tangential quadrilateral [3 marks]

HINTS/SOLUTION	MARKS
Let the circle touch AB, BC, CD and DA at P, Q, R and S respectively.	½
Tangents from an external point are equal: $AP = AS, BP = BQ, CQ = CR, DR = DS.$	1½
$AB + CD = (AP + PB) + (CR + RD)$ $= (AS + BQ) + (CQ + DS)$ $= (AS + SD) + (BQ + QC)$ $= AD + BC.$ Hence proved.	1

Q32. Trigonometric identity [3 marks]

HINTS/SOLUTION	MARKS
$\text{LHS} = (\text{cosec}\theta - \cot\theta)^2 = \left[\frac{1 - \cos\theta}{\sin\theta}\right]^2.$	1
$= \frac{(1 - \cos\theta)^2}{\sin^2\theta} = \frac{(1 - \cos\theta)^2}{1 - \cos^2\theta}.$	1
$= \frac{\mathbf{(1 - \cos\theta)^2}}{\mathbf{[(1 - \cos\theta)(1 + \cos\theta)]}} = \mathbf{\frac{1 - \cos\theta}{1 + \cos\theta}}$ $=$ **RHS. Hence proved.**	1

Q. 33 sugar syrup in 45 gulab jamuns [3 marks]

HINTS SOLUTION	MARKS
Radius r $= \frac{2.8}{2} = 1.4$ cm. Cylindrical length h $= 5 - 2(1.4) = 2.2$ cm.	1
Volume of one $= \pi r^2 h + \left(\frac{4}{3}\right)\pi r^3$ $= \pi\, 1.4^2 \left(2.2 + \left(\frac{4}{3}\right) \times 1.4\right) = \pi\, 1.4^2 \times 2.87$ cm^3.	1
$\therefore$ Syrup in 45 $= 30\% \times 45 \times \pi\, 1.4^2 \times 2.87$ ≈ 338.0 cm^3 (≈ 338 mL).	1

Q34. Mean number of plants [3 marks]

HINTS / SOLUTION	MARKS
Class marks x_i: 1,3,5,7,9,11,13; frequencies f_i: 1,2,1,5,6,2,3.	1
$f_i x_i$: 1,6,5,35,54,22,39; $\Sigma f_i = 20, \Sigma f_i x_i = 162$.	1
$\therefore$ Mean $= \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{162}{20} =$ 8.1 plants. Direct method is convenient because class marks and frequencies are small.	1

Q35. Modal lifetime [3 marks]

HINTS/SOLUTION	MARKS
Highest frequency is 61 in class 60–80; modal class = 60–80. $l = 60, h = 20, f_1 = 61, f_0 = 52, f_2 = 38.$	1
$Mode = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h.$ $= 60 + \left[\frac{61 - 52}{122 - 52 - 38}\right] \times 20$	1
$\therefore$ Mode $= 60 + \frac{180}{32} =$ 65.625 hours.	1

SECTION D — 4 MARKS EACH ($2 \times 4 = 8$)

Q36. Tower and canal width [4 marks]

HINTS/SOLUTION	MARKS
Let BC $= x$ m (canal width), AB = h m (tower height). Points D, C, B are collinear and DC = 20 m.	1
In $\Delta ABC, \tan 60° = \frac{AB}{BC} \Rightarrow h = \sqrt{3x}$. (i)	1
In $\Delta ABD, \tan 30° = \frac{AB}{BD}$ $\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x + 20}$ $\Rightarrow h = \frac{x + 20}{\sqrt{3}}$. (ii)	1
From(i)&(ii) $\sqrt{3x} = \frac{x + 20}{\sqrt{3}}$ $\Rightarrow 3x = x + 20$	1

$\Rightarrow\ x\ =\ 10$ m; h $=\ 10\sqrt{3}$m. $\therefore$ Canal width $=\ 10$ m; tower height $=\ 10\sqrt{3}$m	

Q37. Grazing area [4 marks]

HINTS/SOLUTION	MARKS
The horse is tied at a corner. Since both rope lengths are less than 15 m, accessible region is a quarter circle. Area $=\left(\frac{1}{4}\right)\pi r^2$.	1
(i)For r $=\ 5$ m, area $=\left(\frac{1}{4}\right)\times 3.14\times 25\ =\ 19.625\ m^2$.	1
For r $=\ 10$ m, area $=\left(\frac{1}{4}\right)\times 3.14\times 100\ =\ 78.5\ m^2$.	1
(ii) $\therefore$ Increase $=\ 78.5\ -\ 19.625\ =\ 58.875\ m^2$.	1